CHAPTER 2

Polar Covalent Bonds; Acids and Bases



FIGURE 2.1 The opium poppy is the source of morphine, one of the first "vegetable alkali," or *alkaloids*, to be isolated. (credit: modification of work "*Papaver somniferum*" by Liz West/Flickr, CC BY 2.0)

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WHY THIS CHAPTER? Understanding organic chemistry means knowing not just what happens but also why and how it happens at the molecular level. In this chapter, we'll look at some of the ways that chemists describe and account for chemical reactivity, thereby providing a foundation to understand the specific reactions discussed in subsequent chapters. Topics such as bond polarity, the acid–base behavior of molecules, and hydrogen-bonding are a particularly important part of that foundation.

We saw in the previous chapter how covalent bonds between atoms are described, and we looked at the valence

bond model, which uses hybrid orbitals to account for the observed shapes of organic molecules. Before going on to a systematic study of organic chemistry, however, we still need to review a few fundamental topics. In particular, we need to look more closely at how electrons are distributed in covalent bonds and at some of the consequences that arise when the electrons in a bond are not shared equally between atoms.

2.1 Polar Covalent Bonds and Electronegativity

Up to this point, we've treated chemical bonds as either ionic or covalent. The bond in sodium chloride, for instance, is ionic. Sodium transfers an electron to chlorine to produce Na^+ and Cl^- ions, which are held together in the solid by electrostatic attractions between unlike charges. The C–C bond in ethane, however, is covalent. The two bonding electrons are shared equally by the two equivalent carbon atoms, resulting in a symmetrical electron distribution in the bond. Most bonds, however, are neither fully ionic nor fully covalent but are somewhere between the two extremes. Such bonds are called **polar covalent bonds**, meaning that the bonding electrons are attracted more strongly by one atom than the other so that the electron distribution between atoms is not symmetrical (FIGURE 2.2).

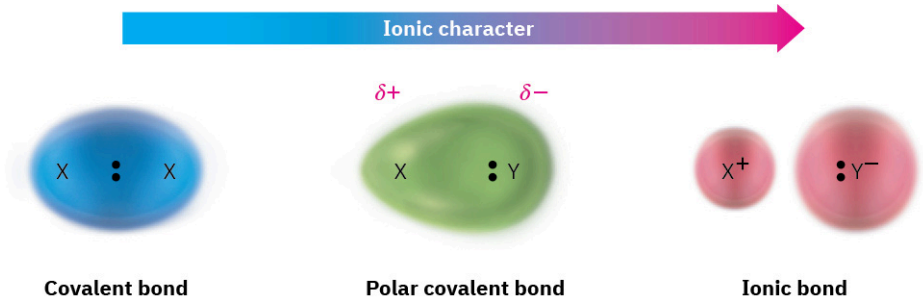


FIGURE 2.2 The continuum in bonding from covalent to ionic is a result of an unequal distribution of bonding electrons between atoms. The symbol δ (lowercase Greek letter delta) means *partial* charge, either partial positive (δ+) for the electron-poor atom or partial negative (δ–) for the electron-rich atom.

Bond polarity is due to differences in **electronegativity (EN)**, the intrinsic ability of an atom to attract the shared electrons in a covalent bond. As shown in FIGURE 2.3, electronegativities are based on an arbitrary scale, with fluorine the most electronegative (EN = 4.0) and cesium the least (EN = 0.7). Metals on the left side of the periodic table attract electrons weakly and have lower electronegativities, while oxygen, nitrogen, and halogens on the right side of the periodic table attract electrons strongly and have higher electronegativities. Carbon, the most important element in organic compounds, has an intermediate electronegativity value of 2.5.

H 2.1																	He
Li 1.0	Be 1.6											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	Ar
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.9	Ni 1.9	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe
Cs 0.7	Ba 0.9	La 1.0	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.9	Bi 1.9	Po 2.0	At 2.1	Rn

FIGURE 2.3 Electronegativity values and trends. Electronegativity generally increases from left to right across the periodic table and decreases from top to bottom. The values are on an arbitrary scale, with F = 4.0 and Cs = 0.7. Elements in **red** are the most electronegative, those in **yellow** are medium, and those in **green** are the least electronegative.

As a rough guide, bonds between atoms whose electronegativities differ by less than 0.5 are nonpolar covalent, bonds between atoms whose electronegativities differ by 0.5 to 2 are polar covalent, and bonds between atoms whose electronegativities differ by more than 2 are largely ionic. Carbon–hydrogen bonds, for example, are relatively nonpolar because carbon (EN = 2.5) and hydrogen (EN = 2.1) have similar electronegativities. Bonds between carbon and more electronegative elements such as oxygen (EN = 3.5) and nitrogen (EN = 3.0), by contrast, are polarized so that the bonding electrons are drawn away from carbon toward the electronegative atom. This leaves carbon with a partial positive charge, δ+, and the electronegative atom with a partial negative charge, δ–. An example is the C–O bond in methanol, CH_3OH (FIGURE 2.4a). Bonds between carbon and less

electronegative elements are polarized so that carbon bears a partial negative charge and the other atom bears a partial positive charge. An example is the C–Li bond in methyllithium, CH_3Li (**FIGURE 2.4**b).

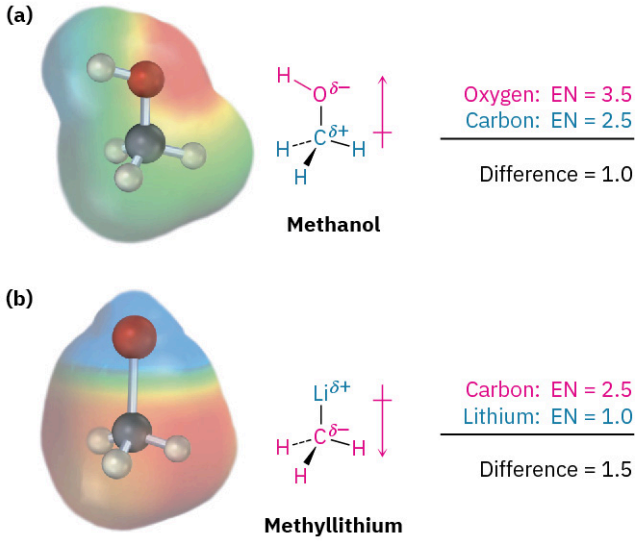


FIGURE 2.4 Polar covalent bonds. (a) Methanol, CH_3OH, has a polar covalent C–O bond, and **(b)** methyllithium, CH_3Li, has a polar covalent C–Li bond. The computer-generated representations, called electrostatic potential maps, use color to show calculated charge distributions, ranging from **red (electron-rich; *δ*–)** to **blue (electron-poor; *δ*+).**

Note in the representations of methanol and methyllithium in **FIGURE 2.4** that a crossed arrow ⟊⟶ is used to indicate the direction of bond polarity. By convention, electrons are displaced in the direction of the arrow. The tail of the arrow (which looks like a plus sign) is electron-poor (*δ*+), and the head of the arrow is electron-rich (*δ*–).

Note also in **FIGURE 2.4** that calculated charge distributions in molecules can be displayed visually with what are called **electrostatic potential maps**, which use color to indicate electron-rich (red; *δ*–) and electron-poor (blue; *δ*+) regions. In methanol, oxygen carries a partial negative charge and is colored red, while the carbon and hydrogen atoms carry partial positive charges and are colored blue-green. In methyllithium, lithium carries a partial positive charge (blue), while carbon and the hydrogen atoms carry partial negative charges (red). Electrostatic potential maps are useful because they show at a glance the electron-rich and electron-poor atoms in molecules. We'll make frequent use of these maps throughout the text and will see many examples of how electronic structure correlates with chemical reactivity.

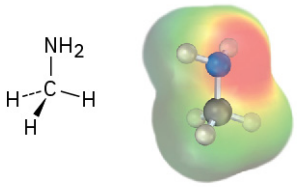
When speaking of an atom's ability to polarize a bond, we often use the term *inductive effect.* An **inductive effect** is simply the shifting of electrons in a *σ* bond in response to the electronegativity of nearby atoms. Metals, such as lithium and magnesium, inductively donate electrons, whereas reactive nonmetals, such as oxygen and nitrogen, inductively withdraw electrons. Inductive effects play a major role in understanding chemical reactivity, and we'll use them many times throughout this text to explain a variety of chemical observations.

PROBLEM 2-1 Which element in each of the following pairs is more electronegative?
(a) Li or H **(b)** B or Br **(c)** Cl or I **(d)** C or H

PROBLEM 2-2 Use the *δ*+/*δ*– convention to indicate the direction of expected polarity for each of the bonds indicated.
(a) $H_3C–Cl$ **(b)** $H_3C–NH_2$ **(c)** $H_2N–H$ **(d)** $H_3C–SH$ **(e)** $H_3C–MgBr$ **(f)** $H_3C–F$

PROBLEM 2-3 Use the electronegativity values shown in Figure 2.3 to rank the following bonds from least polar to most polar: $H_3C–Li$, $H_3C–K$, $H_3C–F$, $H_3C–MgBr$, $H_3C–OH$

PROBLEM 2-4 Look at the following electrostatic potential map of methylamine, a substance responsible for the odor of rotting fish, and tell the direction of polarization of the C–N bond:



Methylamine

2.2 Polar Covalent Bonds and Dipole Moments

Just as individual bonds are often polar, molecules as a whole are often polar as well. Molecular polarity results from the vector summation of all individual bond polarities and lone-pair contributions in the molecule. As a practical matter, strongly polar substances are often soluble in polar solvents like water, whereas less polar substances are insoluble in water.

Net polarity is measured by a quantity called the *dipole moment* and can be thought of in the following way: assume that there is a center of mass of all positive charges (nuclei) in a molecule and a center of mass of all negative charges (electrons). If these two centers don't coincide, then the molecule has a net polarity.

The **dipole moment**, μ (lowercase Greek letter mu), is defined as the magnitude of the charge Q at either end of the molecular dipole times the distance r between the charges, $\mu = Q \times r$. Dipole moments are expressed in *debyes* (D), where 1 D = 3.336×10^{-30} coulomb meters (C · m) in SI units. For example, the unit charge on an electron is 1.60×10^{-19} C. Thus, if one positive charge and one negative charge are separated by 100 pm (a bit less than the length of a typical covalent bond), the dipole moment is 1.60×10^{-29} C · m, or 4.80 D.

$$\mu = Q \times r$$

$$\mu = (1.60 \times 10^{-19}\ \text{C})(100 \times 10^{-12}\ \text{m})\left(\frac{1\ \text{D}}{3.336 \times 10^{-30}\ \text{C} \cdot \text{m}}\right) = 4.80\ \text{D}$$

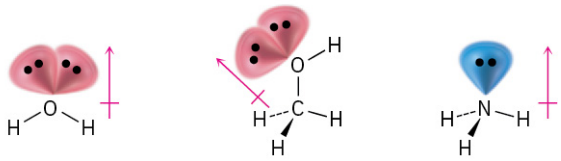
Dipole moments for some common substances are given in TABLE 2.1. Of the compounds shown in the table, sodium chloride has the largest dipole moment (9.00 D) because it is ionic. Even small molecules like water (μ = 1.85 D), methanol (CH_3OH; μ = 1.70 D), and ammonia (μ = 1.47 D), have substantial dipole moments, however, both because they contain strongly electronegative atoms (oxygen and nitrogen) and because all three molecules have lone-pair electrons. The lone-pair electrons on oxygen and nitrogen stick out into space away from the positively charged nuclei, giving rise to a considerable charge separation and making a large contribution to the dipole moment.

TABLE 2.1 Dipole Moments of Some Compounds

Compound	Dipole moment (D)	Compound	Dipole moment (D)
NaCl	9.00	NH_3	1.47
CH_2O	2.33	CH_3NH_2	1.31
CH_3Cl	1.87	CO_2	0
H_2O	1.85	CH_4	0
CH_3OH	1.70	CH_3CH_3	0

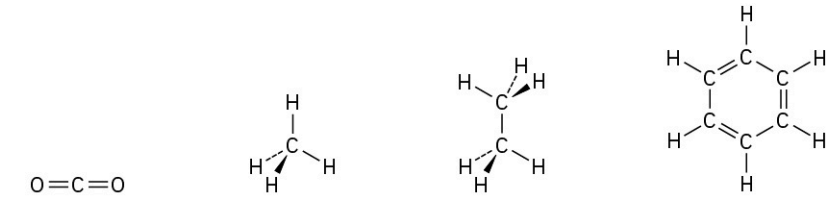
TABLE 2.1 Dipole Moments of Some Compounds

Compound	Dipole moment (D)	Compound	Dipole moment (D)
CH_3CO_2H	1.70	 Benzene	0
CH_3SH	1.52		



Water (μ = 1.85 D) **Methanol** (μ = 1.70 D) **Ammonia** (μ = 1.47 D)

In contrast with water, methanol, and ammonia, molecules such as carbon dioxide, methane, ethane, and benzene have zero dipole moments. Because of the symmetrical structures of these molecules, the individual bond polarities and lone-pair contributions exactly cancel.



Carbon dioxide (μ = 0) **Methane** (μ = 0) **Ethane** (μ = 0) **Benzene** (μ = 0)

WORKED EXAMPLE 2.1

Predicting the Direction of a Dipole Moment

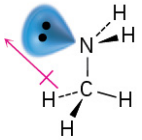
Make a three-dimensional drawing of methylamine, CH_3NH_2, and show the direction of its dipole moment (μ = 1.31).

Strategy

Look for any lone-pair electrons, and identify any atom with an electronegativity substantially different from that of carbon. (Usually, this means O, N, F, Cl, or Br.) Electron density will be displaced in the general direction of the electronegative atoms and the lone pairs.

Solution

Methylamine contains an electronegative nitrogen atom with a lone pair of electrons. The dipole moment thus points generally from $-CH_3$ toward the lone pair.



Methylamine (μ = 1.31)

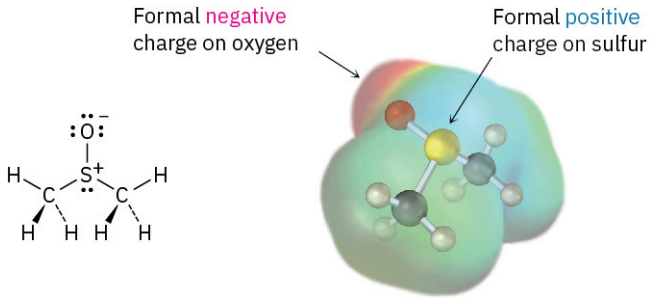
PROBLEM 2-5 Ethylene glycol, $HOCH_2CH_2OH$, may look nonpolar when drawn, but an internal hydrogen bond between the two –OH groups results in a dipole moment. Explain.

PROBLEM 2-6 Make three-dimensional drawings of the following molecules, and predict whether each has a dipole moment. If you expect a dipole moment, show its direction.

(a) $H_2C{=}CH_2$ **(b)** $CHCl_3$ **(c)** CH_2Cl_2 **(d)** $H_2C{=}CCl_2$

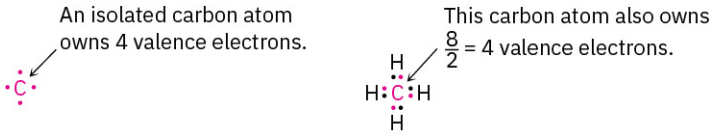
2.3 Formal Charges

Closely related to the ideas of bond polarity and dipole moment is the assignment of *formal charges* to specific atoms within a molecule, particularly atoms that have an apparently "abnormal" number of bonds. Look at dimethyl sulfoxide (CH_3SOCH_3), for instance, a solvent commonly used for preserving biological cell lines at low temperature. The sulfur atom in dimethyl sulfoxide has three bonds rather than the usual two and has a formal positive charge. The oxygen atom, by contrast, has one bond rather than the usual two and has a formal negative charge. Note that an electrostatic potential map of dimethyl sulfoxide shows the oxygen as negative (red) and the sulfur as relatively positive (blue), in accordance with the formal charges.

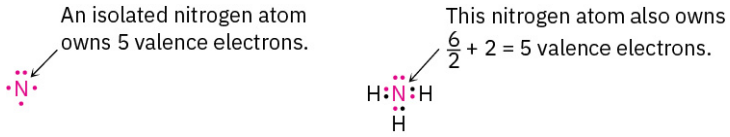


Dimethyl sulfoxide

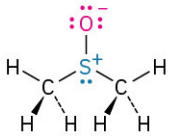
Formal charges, as the name suggests, are a formalism and don't imply the presence of actual ionic charges in a molecule. Instead, they're a device for electron "bookkeeping" and can be thought of in the following way: A typical covalent bond is formed when each atom donates one electron. Although the bonding electrons are shared by both atoms, each atom can still be considered to "own" one electron for bookkeeping purposes. In methane, for instance, the carbon atom owns one electron in each of the four C–H bonds. Because a neutral, isolated carbon atom has four valence electrons, and because the carbon atom in methane still owns four, the methane carbon atom is neutral and has no formal charge.



The same is true for the nitrogen atom in ammonia, which has three covalent N–H bonds and two nonbonding electrons (a lone pair). Atomic nitrogen has five valence electrons, and the ammonia nitrogen also has five—one in each of three shared N–H bonds plus two in the lone pair. Thus, the nitrogen atom in ammonia has no formal charge.



The situation is different in dimethyl sulfoxide. Atomic sulfur has six valence electrons, but the dimethyl sulfoxide sulfur owns only five—one in each of the two S–C single bonds, one in the S–O single bond, and two in a lone pair. Thus, the sulfur atom has formally lost an electron and therefore has a positive formal charge. A similar calculation for the oxygen atom shows that it has formally gained an electron and has a negative charge. Atomic oxygen has six valence electrons, but the oxygen in dimethyl sulfoxide has seven—one in the O–S bond and two in each of three lone pairs. Thus, the oxygen has formally gained an electron and has a negative formal charge.



For sulfur:

Sulfur valence electrons = 6
Sulfur bonding electrons = 6
Sulfur nonbonding electrons = 2

Formal charge = 6 − 6/2 − 2 = +1

For oxygen:

Oxygen valence electrons = 6
Oxygen bonding electrons = 2
Oxygen nonbonding electrons = 6

Formal charge = 6 − 2/2 − 6 = −1

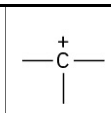
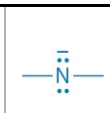
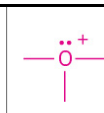
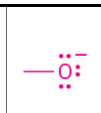
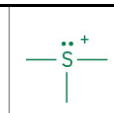
To express the calculations in a general way, the **formal charge** on an atom is equal to the number of valence electrons in a neutral, isolated atom minus the number of electrons owned by that bonded atom in a molecule. The number of electrons in the bonded atom, in turn, is equal to half the number of bonding electrons plus the nonbonding, lone-pair electrons.

$$\textbf{Formal charge} = \left(\begin{array}{c}\text{Number of}\\ \text{valence electrons}\\ \text{in free atom}\end{array}\right) - \left(\begin{array}{c}\text{Number of}\\ \text{valence electrons}\\ \text{in bonded atom}\end{array}\right)$$

$$= \left(\begin{array}{c}\text{Number of}\\ \text{valence electrons}\\ \text{in free atom}\end{array}\right) - \left(\frac{\begin{array}{c}\text{Number of}\\ \text{bonding electrons}\end{array}}{2} + \begin{array}{c}\text{Number of}\\ \text{nonbonding}\\ \text{electrons}\end{array}\right)$$

A summary of commonly encountered formal charges and the bonding situations in which they occur is given in TABLE 2.2. Although only a bookkeeping device, formal charges often give clues about chemical reactivity, so it's helpful to be able to identify and calculate them correctly.

TABLE 2.2 A Summary of Common Formal Charges

Atom	C			N		O		S		P
Structure										
Valence electrons	4	4	4	5	5	6	6	6	6	5
Number of bonds	3	3	3	4	2	3	1	3	1	4
Number of nonbonding electrons	1	0	2	0	4	2	6	2	6	0
Formal charge	0	+1	−1	+1	−1	+1	−1	+1	−1	+1

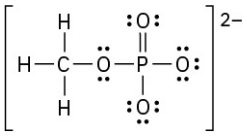
PROBLEM 2-7 Calculate formal charges for the nonhydrogen atoms in the following molecules:

(a) Diazomethane, $H_2C{=}N{=}\ddot{N}:$ **(b)** Acetonitrile oxide, $H_3C{-}C{\equiv}N{-}\ddot{O}:$

(c) Methyl isocyanide, $H_3C{-}N{\equiv}C:$

PROBLEM 2-8 Organic phosphate groups occur commonly in biological molecules. Calculate formal charges on

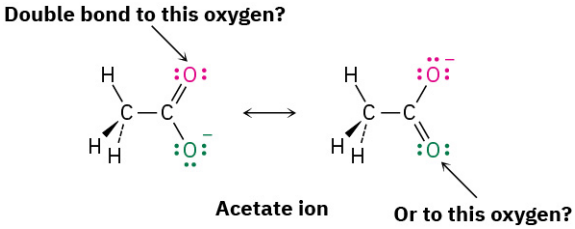
the four O atoms in the methyl phosphate dianion.



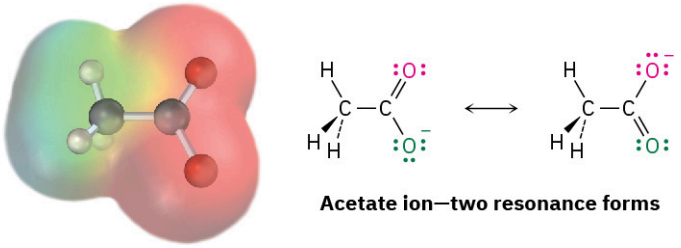
Methyl phosphate dianion

2.4 Resonance

Most substances can be represented unambiguously by the Kekulé line-bond structures we've been using up to this point, but an interesting problem sometimes arises. Look at the acetate ion, for instance. When we draw a line-bond structure for acetate, we need to show a double bond to one oxygen and a single bond to the other. But which oxygen is which? Should we draw a double bond to the "top" oxygen and a single bond to the "bottom" oxygen, or vice versa?



Although the two oxygen atoms in the acetate ion appear different in line-bond structures, experiments show that they are equivalent. Both carbon–oxygen bonds, for instance, are 127 pm in length, midway between the length of a typical C–O single bond (135 pm) and a typical C=O double bond (120 pm). In other words, neither of the two structures for acetate is correct by itself. The true structure is intermediate between the two, and an electrostatic potential map shows that both oxygen atoms share the negative charge and have equal electron densities (red).



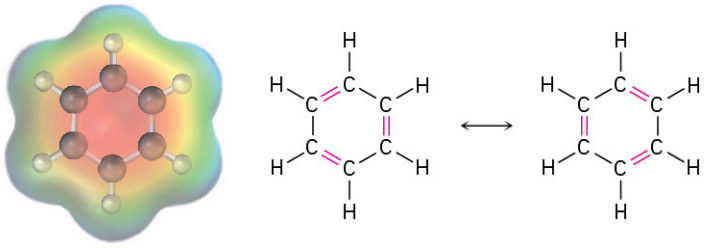
Acetate ion—two resonance forms

The two individual line-bond structures for acetate ion are called **resonance forms**, and their special resonance relationship is indicated by the double-headed arrow between them. The only difference between the two resonance forms is the placement of the π and nonbonding valence electrons. The atoms themselves occupy exactly the same place in both resonance forms, the connections between atoms are the same, and the three-dimensional shapes of the resonance forms are the same.

A good way to think about resonance forms is to realize that a substance like the acetate ion is the same as any other. Acetate doesn't jump back and forth between two resonance forms, spending part of the time looking like one and part of the time looking like the other. Rather, acetate has a single unchanging structure that we say is a **resonance hybrid** of the two individual forms and has characteristics of both. The only "problem" with acetate is that we can't draw it accurately using a familiar line-bond structure—line-bond structures just don't work for resonance hybrids. The difficulty, however, is with the *representation* of acetate on paper, not with acetate itself.

Resonance is a very useful concept that we'll return to on numerous occasions throughout the rest of this book. We'll see in Chapter 15, for instance, that the six carbon–carbon bonds in aromatic compounds, such as benzene, are equivalent and that benzene is best represented as a hybrid of two resonance forms. Although each individual resonance form seems to imply that benzene has alternating single and double bonds, neither form is correct by itself. The true benzene structure is a hybrid of the two individual forms, and all six carbon–carbon bonds are equivalent. This symmetrical distribution of electrons around the molecule is evident in an

electrostatic potential map.



Benzene (two resonance forms)

2.5 Rules for Resonance Forms

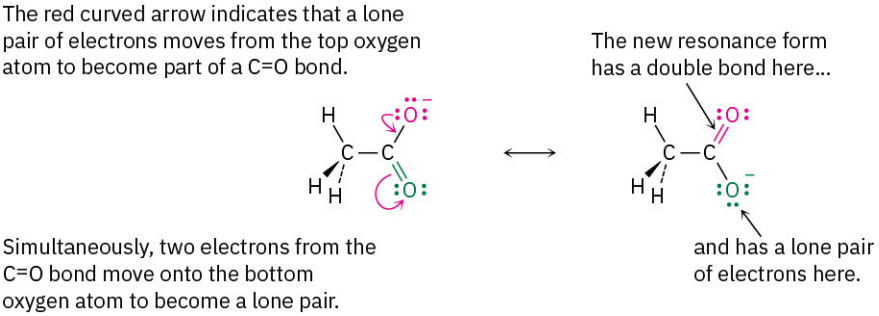
When first dealing with resonance forms, it's useful to have a set of guidelines that describe how to draw and interpret them. The following rules should be helpful:

RULE 1

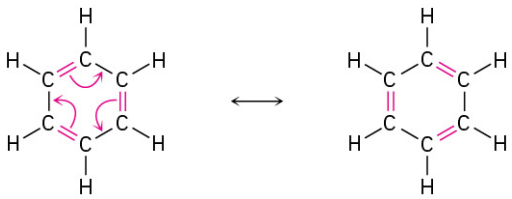
Individual resonance forms are imaginary, not real. The real structure is a composite, or resonance hybrid, of the different forms. Species such as the acetate ion and benzene are no different from any other. They have single, unchanging structures, and they don't switch back and forth between resonance forms. The only difference between these and other substances is in the way they are represented in drawings.

RULE 2

Resonance forms differ only in the placement of their π or nonbonding electrons. Neither the position nor the hybridization of any atom changes from one resonance form to another. In the acetate ion, for instance, the carbon atom is sp^2-hybridized and the oxygen atoms remain in exactly the same place in both resonance forms. Only the positions of the π electrons in the C=O bond and the lone-pair electrons on oxygen differ from one form to another. This movement of electrons from one resonance structure to another can be indicated with curved arrows. *A curved arrow always indicates the movement of electrons, not the movement of atoms.* An arrow shows that a pair of electrons moves *from* the atom or bond at the tail of the arrow *to* the atom or bond at the head of the arrow.



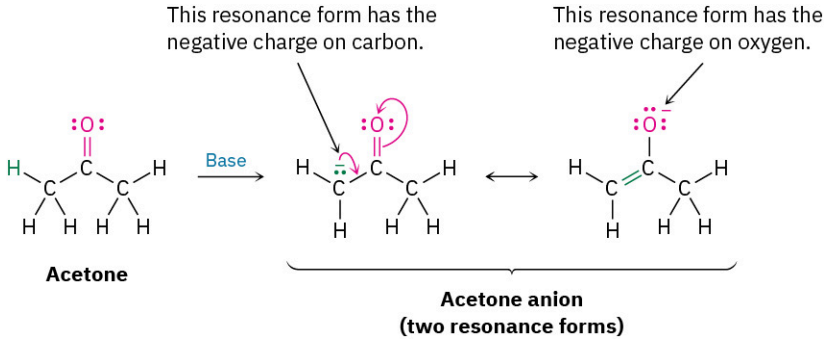
The situation with benzene is similar to that with acetate. The π electrons in the double bonds move, as shown with curved arrows, but the carbon and hydrogen atoms remain in place.



RULE 3

Different resonance forms of a substance don't have to be equivalent. As an example, we'll see in Chapter 22 that a compound such as acetone, which contains a C=O bond, can be converted into its anion by reaction with a strong base. The resultant anion has two resonance forms. One form contains a carbon–*oxygen* double bond and has a negative charge on *carbon;* the other contains a carbon–*carbon* double bond and has a negative charge on *oxygen.* Even though the two resonance forms aren't equivalent, both contribute to the overall

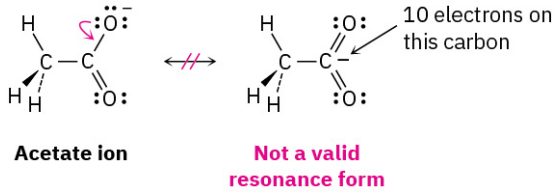
resonance hybrid.



When two resonance forms are nonequivalent, the actual structure of the resonance hybrid resembles the more stable form. Thus, we might expect the true structure of the acetone anion to be more like that of the form that places the negative charge on the electronegative oxygen atom rather than on the carbon.

RULE 4

Resonance forms obey normal rules of valency. A resonance form is like any other structure: the octet rule still applies to second-row, main-group atoms. For example, one of the following structures for the acetate ion is not a valid resonance form because the carbon atom has five bonds and ten valence electrons:

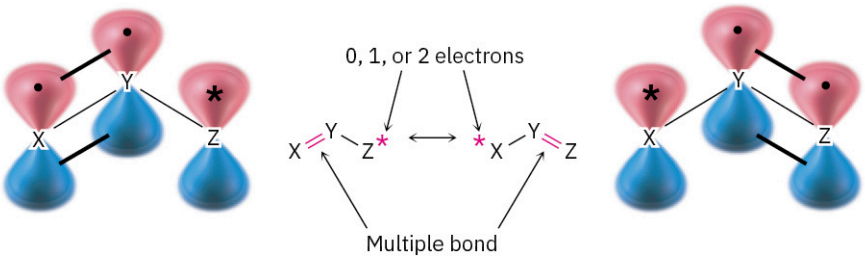


RULE 5

The resonance hybrid is more stable than any individual resonance form. In other words, resonance leads to stability. Generally speaking, the larger the number of resonance forms a substance has, the more stable the substance is, because its electrons are spread out over a larger part of the molecule and are closer to more nuclei. We'll see in Chapter 15, for instance, that a benzene ring is more stable because of resonance than might otherwise be expected.

2.6 Drawing Resonance Forms

Look back at the resonance forms of the acetate ion and the acetone anion shown in the previous section. The pattern seen there is a common one that leads to a useful technique for drawing resonance forms. In general, any three-atom grouping with a *p* orbital on each atom has two resonance forms:



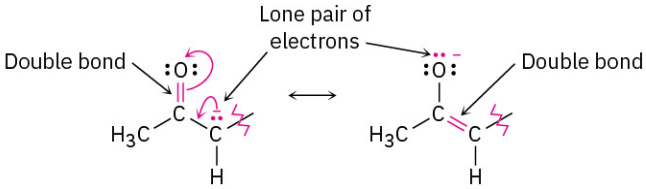
The atoms X, Y, and Z in the general structure might be C, N, O, P, S, or others, and the asterisk (*) might mean that the *p* orbital on atom Z is vacant, that it contains a single electron, or that it contains a lone pair of electrons. The two resonance forms differ simply by an exchange in position of the multiple bond and the asterisk from one end of the three-atom grouping to the other.

By learning to recognize such three-atom groupings within larger structures, resonance forms can be systematically generated. Look, for instance, at the anion produced when H^+ is removed from 2,4-pentanedione by reaction with a base. How many resonance structures does the resultant anion have?

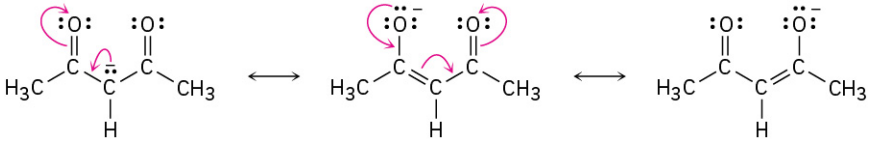


2,4-Pentanedione

The 2,4-pentanedione anion has a lone pair of electrons and a formal negative charge on the central carbon atom, next to a C=O bond on the left. The O=C–C:⁻ grouping is a typical one for which two resonance structures can be drawn.



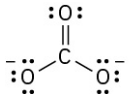
Just as there is a C=O bond to the left of the lone pair, there is a second C=O bond to the right. Thus, we can draw a total of three resonance structures for the 2,4-pentanedione anion.



WORKED EXAMPLE 2.2

Drawing Resonance Forms for an Anion

Draw three resonance structures for the carbonate ion, CO_3^{2-}.



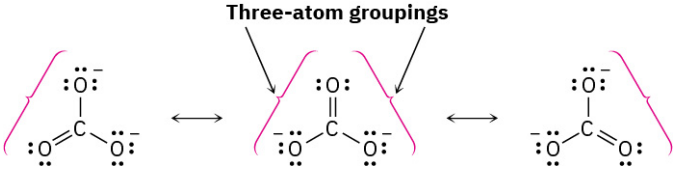
Carbonate ion

Strategy

Look for three-atom groupings that contain a multiple bond next to an atom with a *p* orbital. Then exchange the positions of the multiple bond and the electrons in the *p* orbital. In the carbonate ion, each singly bonded oxygen atom with three lone pairs and a negative charge is adjacent to the C=O double bond, giving the grouping Ö=C–Ö:⁻.

Solution

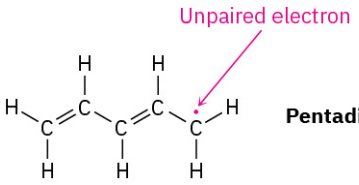
Exchanging the position of the double bond and an electron lone pair in each grouping generates three resonance structures.



WORKED EXAMPLE 2.3

Drawing Resonance Forms for a Radical

Draw three resonance forms for the pentadienyl radical, where a **radical** is a substance that contains a single, unpaired electron in one of its orbitals, denoted by a dot (·).



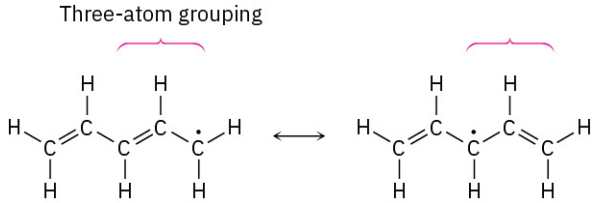
Pentadienyl radical

Strategy

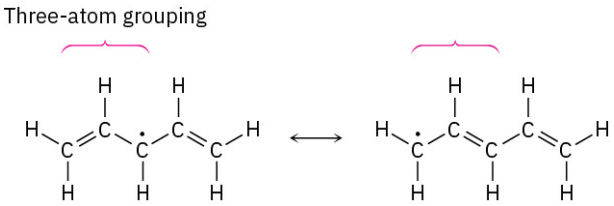
Find the three-atom groupings that contain a multiple bond next to an atom with a *p* orbital.

Solution

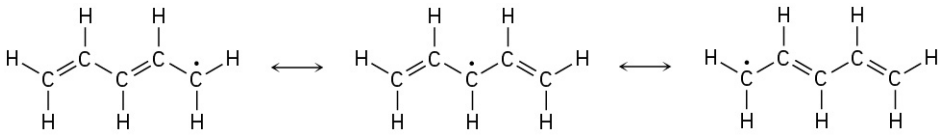
The unpaired electron is on a carbon atom next to a C=C bond, giving a typical three-atom grouping that has two resonance forms.



In the second resonance form, the unpaired electron is next to another double bond, giving another three-atom grouping and leading to another resonance form.

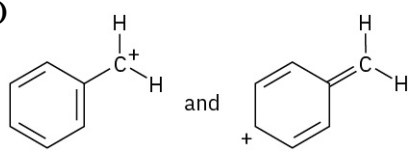


Thus, the three resonance forms for the pentadienyl radical are:

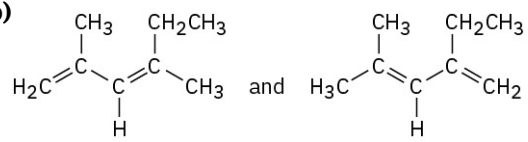


PROBLEM 2-9 Which of the following pairs of structures represent resonance forms, and which do not? Explain.

(a)



(b)



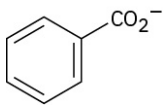
PROBLEM 2-10 Draw the indicated number of resonance forms for each of the following substances:

(a) The methyl phosphate anion, $CH_3OPO_3^{2-}$ (3)

(b) The nitrate anion, NO_3^- (3)

(c) The allyl cation, $H_2C{=}CH{-}CH_2^+$ (2)

(d) The benzoate anion (2)

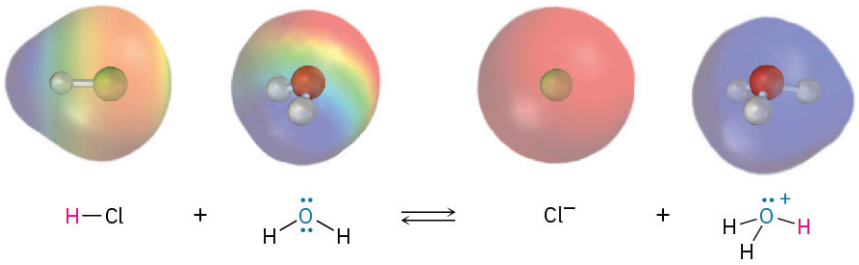


2.7 Acids and Bases: The Brønsted–Lowry Definition

Perhaps the most important of all concepts related to electronegativity and polarity is that of *acidity* and

basicity. We'll soon see, in fact, that the acid–base behavior of organic molecules explains much of their chemistry. You may recall from a course in general chemistry that two definitions of acidity are frequently used: the *Brønsted–Lowry definition* and the *Lewis definition.* We'll look at the Brønsted–Lowry definition in this and the following three sections and then discuss the Lewis definition in **Section 2.11**.

A **Brønsted–Lowry acid** is a substance that donates a hydrogen ion, H^+, and a **Brønsted–Lowry base** is a substance that accepts a hydrogen ion. (The name *proton* is often used as a synonym for H^+ because loss of the valence electron from a neutral hydrogen atom leaves only the hydrogen nucleus—a proton.) When gaseous hydrogen chloride dissolves in water, for example, a polar HCl molecule acts as an acid and donates a proton, while a water molecule acts as a base and accepts the proton, yielding chloride anion (Cl^-) and hydronium cation (H_3O^+). This and other acid–base reactions are reversible, so we'll write them with double, forward-and-backward arrows.

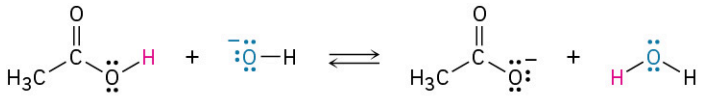


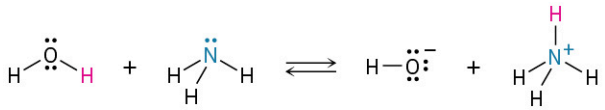
Chloride ion, the product that results when the acid HCl loses a proton, is called the **conjugate base** of the acid, and hydronium ion, the product that results when the base H_2O gains a proton, is called the **conjugate acid** of the base. Other common mineral acids such as H_2SO_4 and HNO_3 behave similarly, as do organic acids such as acetic acid, CH_3CO_2H.

In a general sense,



For example:





Notice that water can act either as an acid or as a base, depending on the circumstances. In its reaction with HCl, water is a base that accepts a proton to give the hydronium ion, H_3O^+. In its reaction with ammonia (NH_3), however, water is an acid that donates a proton to give ammonium ion (NH_4^+) and hydroxide ion, HO^-.

PROBLEM 2-11 Nitric acid (HNO_3) reacts with ammonia (NH_3) to yield ammonium nitrate. Write the reaction, and identify the acid, the base, the conjugate acid product, and the conjugate base product.

2.8 Acid and Base Strength

Different acids differ in their ability to donate H^+. Stronger acids, such as HCl, react almost completely with water, whereas weaker acids, such as acetic acid (CH_3CO_2H), react only slightly. The exact strength of a given acid HA in water solution is described using the **acidity constant (K_a)** for the acid-dissociation equilibrium. Recall from general chemistry that the concentration of solvent is ignored in the equilibrium expression and that brackets [ ] around a substance refer to the concentration of the enclosed species in moles per liter.

$$HA + H_2O \rightleftharpoons A^- + H_3O^+$$

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

Stronger acids have their equilibria toward the right and thus have larger acidity constants, whereas weaker acids have their equilibria toward the left and have smaller acidity constants. The range of K_a values for different acids is enormous, running from about 10^{15} for the strongest acids to about 10^{-60} for the weakest. Common inorganic acids such as H_2SO_4, HNO_3, and HCl have K_a's in the range of 10^2 to 10^9, while organic acids generally have K_a's in the range of 10^{-5} to 10^{-15}. As you gain experience, you'll develop a rough feeling for which acids are "strong" and which are "weak" (always remembering that the terms are relative, not absolute).

Acid strengths are normally expressed using pK_a values rather than K_a values, where the **pK_a** is the negative common logarithm of the K_a:

$$pK_a = -\log K_a$$

A stronger acid (larger K_a) has a smaller pK_a, and a weaker acid (smaller K_a) has a larger pK_a. TABLE 2.3 lists the pK_a's of some common acids in order of their strength, and a more comprehensive table is given in Appendix B.

Notice that the pK_a value shown in TABLE 2.3 for water is 15.74, which results from the following calculation. Because water is both the acid and the solvent, the equilibrium expression is

$$\underset{\text{(acid)}}{H_2O} + \underset{\text{(solvent)}}{H_2O} \rightleftharpoons OH^- + H_3O^+$$

$$K_a = \frac{[H_3O^+][A^-]}{[HA]} = \frac{[H_3O^+][OH^-]}{[H_2O]} = \frac{[1.0 \times 10^{-7}][1.0 \times 10^{-7}]}{[55.4]} = 1.8 \times 10^{-16}$$

$$pK_a = 15.74$$

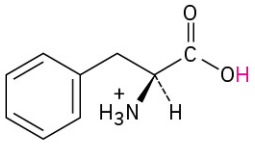
The numerator in this expression is the so-called ion-product constant for water, $K_w = [H_3O^+][OH^-] = 1.00 \times 10^{-14}$, and the denominator is the molar concentration of pure water, $[H_2O] = 55.4$ M at 25 °C. The calculation is artificial in that the concentration of "solvent" water is ignored while the concentration of "acid" water is not, but it is nevertheless useful for making a comparison of water with other weak acids on a similar footing.

Notice also in TABLE 2.3 that there is an inverse relationship between the acid strength of an acid and the base strength of its conjugate base. A strong acid has a weak conjugate base, and a weak acid has a strong conjugate base. To understand this inverse relationship, think about what is happening to the acidic hydrogen in an acid–base reaction. A strong acid is one that loses H^+ easily, meaning that its conjugate base holds the H^+ weakly and is therefore a weak base. A weak acid is one that loses H^+ with difficulty, meaning that its conjugate base holds the proton tightly and is therefore a strong base. The fact that HCl is a strong acid, for example, means that Cl^- does not hold H^+ tightly and is thus a weak base. Water, on the other hand, is a weak acid, meaning that OH^- holds H^+ tightly and is a strong base.

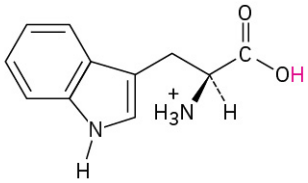
TABLE 2.3 Relative Strengths of Some Common Acids and Their Conjugate Bases

	Acid	Name	pK_a	Conjugate base	Name	
Weaker acid	CH_3CH_2OH	Ethanol	16.00	$CH_3CH_2O^-$	Ethoxide ion	Stronger base
	H_2O	Water	15.74	HO^-	Hydroxide ion	
	HCN	Hydrocyanic acid	9.31	CN^-	Cyanide ion	
	$H_2PO_4^-$	Dihydrogen phosphate ion	7.21	HPO_4^{2-}	Hydrogen phosphate ion	
	CH_3CO_2H	Acetic acid	4.76	$CH_3CO_2^-$	Acetate ion	
	H_3PO_4	Phosphoric acid	2.16	$H_2PO_4^-$	Dihydrogen phosphate ion	
	HNO_3	Nitric acid	−1.3	NO_3^-	Nitrate ion	
Stronger acid	HCl	Hydrochloric acid	−7.0	Cl^-	Chloride ion	Weaker base

PROBLEM 2-12 The amino acid phenylalanine has pK_a = 1.83, and tryptophan has pK_a = 2.83. Which is the stronger acid?



Phenylalanine (pK_a = 1.83)

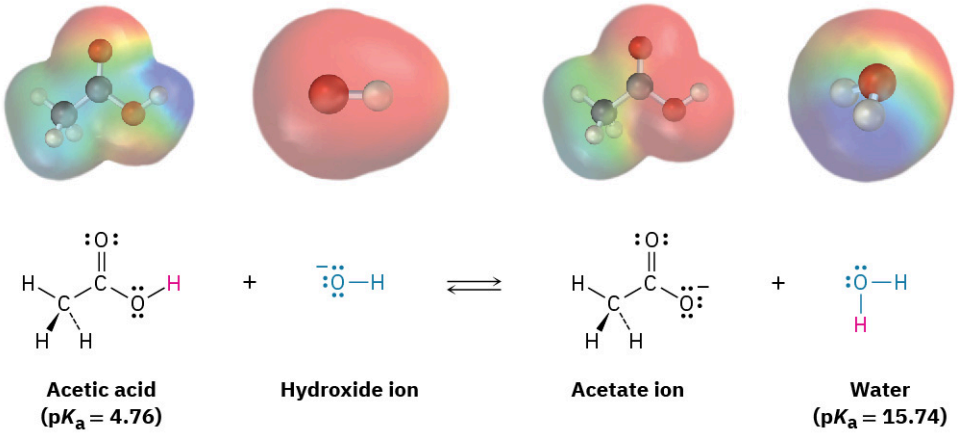


Tryptophan (pK_a = 2.83)

PROBLEM 2-13 Amide ion, H_2N^-, is a much stronger base than hydroxide ion, HO^-. Which is the stronger acid, NH_3 or H_2O? Explain.

2.9 Predicting Acid–Base Reactions from pK_a Values

Compilations of pK_a values like those in TABLE 2.3 and Appendix B are useful for predicting whether a given acid–base reaction will take place, because H^+ will always go *from* the stronger acid *to* the stronger base. That is, an acid will donate a proton to the conjugate base of a weaker acid, and the conjugate base of a weaker acid will remove a proton from a stronger acid. Since water (pK_a = 15.74) is a weaker acid than acetic acid (pK_a = 4.76), for example, hydroxide ion holds a proton more tightly than acetate ion does. Hydroxide ion will therefore react to a large extent with acetic acid, CH_3CO_2H, to yield acetate ion and H_2O.



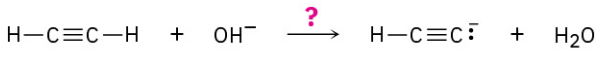
Another way to predict acid–base reactivity is to remember that the product conjugate acid in an acid–base reaction must be weaker and less reactive than the starting acid and the product conjugate base must be weaker and less reactive than the starting base. In the reaction of acetic acid with hydroxide ion, for example, the product conjugate acid (H_2O) is weaker than the starting acid (CH_3CO_2H), and the product conjugate base ($CH_3CO_2^-$) is weaker than the starting base (OH^-).



WORKED EXAMPLE 2.4

Predicting Acid Strengths from pK_a Values

Water has pK_a = 15.74, and acetylene has pK_a = 25. Which is the stronger acid? Does hydroxide ion react to a significant extent with acetylene?



Strategy

In comparing two acids, the one with the lower pK_a is stronger. Thus, water is a stronger acid than acetylene and gives up H^+ more easily.

Solution

Because water is a stronger acid and gives up H^+ more easily than acetylene, the HO^- ion must have less affinity for H^+ than the $HC{\equiv}C{:}^-$ ion. In other words, the anion of acetylene is a stronger base than hydroxide ion, and the reaction will not proceed significantly as written.

WORKED EXAMPLE 2.5

Calculating K_a from pK_a

According to the data in TABLE 2.3, acetic acid has pK_a = 4.76. What is its K_a?

Strategy

Since pK_a is the negative logarithm of K_a, it's necessary to use a calculator with an ANTILOG or INV LOG function. Enter the value of the pK_a (4.76), change the sign (–4.76), and then find the antilog (1.74×10^{-5}).

Solution

$K_a = 1.74 \times 10^{-5}$.

PROBLEM 2-14 Will either of the following reactions take place to a significant extent as written, according to the data in Table 2.3?

(a)



(b)



PROBLEM 2-15 Ammonia, NH_3, has $pK_a \approx 36$, and acetone has $pK_a \approx 19$. Will the following reaction take place to a significant extent?



Acetone

PROBLEM 2-16 What is the K_a of HCN if its $pK_a = 9.31$?

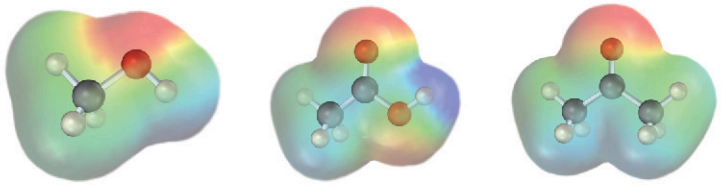
2.10 Organic Acids and Organic Bases

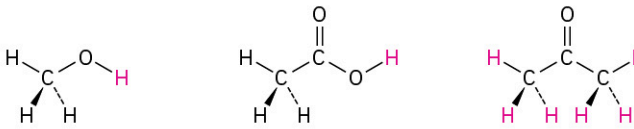
Many of the reactions we'll be seeing in future chapters, including practically all biological reactions, involve organic acids and organic bases. Although it's too early to go into the details of these processes now, you might keep the following generalities in mind:

Organic Acids

Organic acids are characterized by the presence of a positively polarized hydrogen atom (blue in electrostatic potential maps) and are of two main kinds: acids such as methanol and acetic acid that contain a hydrogen atom bonded to an electronegative oxygen atom (O–H) and those such as acetone (Section 2.5) that contain a hydrogen atom bonded to a carbon atom next to a C=O bond (O=C–C–H).

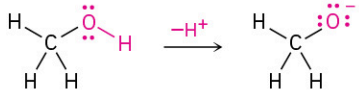
Some organic acids



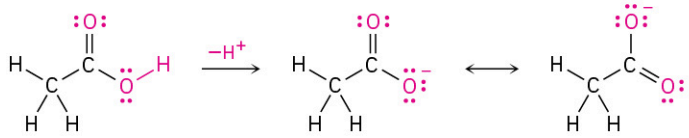


Methanol ($pK_a = 15.54$) **Acetic acid ($pK_a = 4.76$)** **Acetone ($pK_a = 19.3$)**

Methanol contains an O–H bond and is a weak acid, while acetic acid also contains an O–H bond and is a somewhat stronger acid. In both cases, acidity is due to the fact that the conjugate base resulting from loss of H^+ is stabilized by having its negative charge on a strongly electronegative oxygen atom. In addition, the conjugate base of acetic acid is stabilized by resonance (Section 2.4 and Section 2.5).

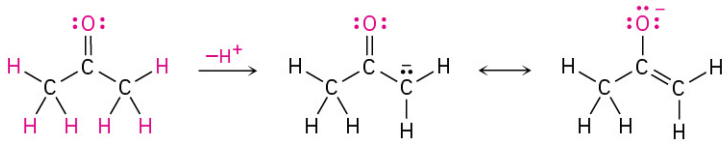


Anion is stabilized by having negative charge on a highly electronegative atom.



Anion is stabilized both by having negative charge on a highly electronegative atom and by resonance.

The acidity of acetone and other compounds with C=O bonds is due to the fact that the conjugate base resulting from loss of H^+ is stabilized by resonance. In addition, one of the resonance forms stabilizes the negative charge by placing it on an electronegative oxygen atom.



Anion is stabilized both by resonance and by having negative charge on a highly electronegative atom.

Electrostatic potential maps of the conjugate bases from methanol, acetic acid, and acetone are shown in **FIGURE 2.5**. As you might expect, all three show a substantial amount of negative charge (red) on oxygen.

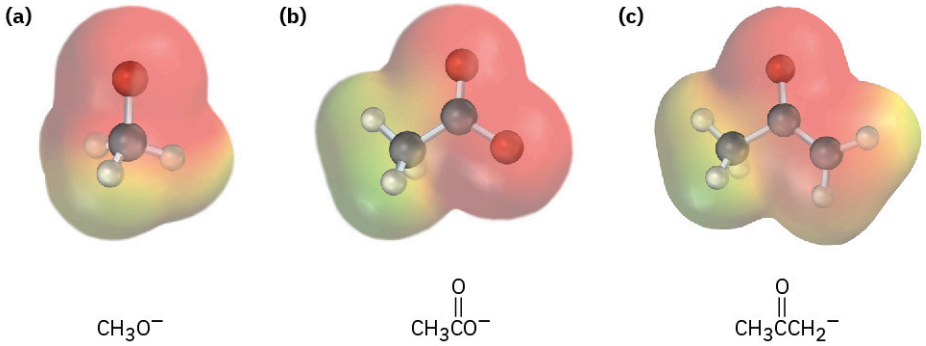
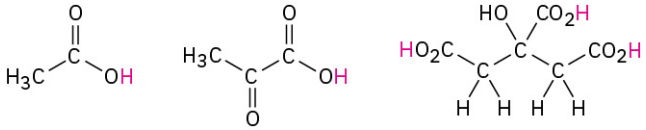


FIGURE 2.5 Electrostatic potential maps of the conjugate bases of (a) methanol, (b) acetic acid, and (c) acetone. The electronegative oxygen atoms stabilize the negative charge in all three.

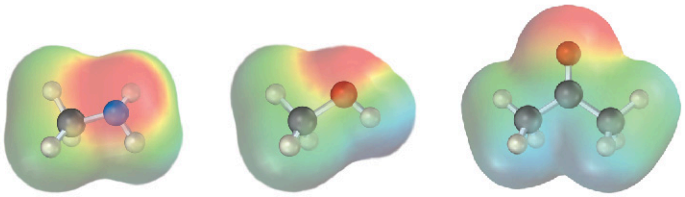
Compounds called *carboxylic acids,* which contain the $-CO_2H$ grouping, occur abundantly in all living organisms and are involved in almost all metabolic pathways. Acetic acid, pyruvic acid, and citric acid are examples. You might note that at the typical pH of 7.3 found within cells, carboxylic acids are usually dissociated and exist as their carboxylate anions, $-CO_2^-$.



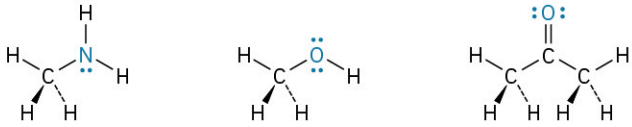
Acetic acid **Pyruvic acid** **Citric acid**

Organic Bases

Organic bases are characterized by the presence of an atom (reddish in electrostatic potential maps) with a lone pair of electrons that can bond to H^+. Nitrogen-containing compounds such as methylamine are the most common organic bases and are involved in almost all metabolic pathways, but oxygen-containing compounds can also act as bases when reacting with a sufficiently strong acid. Note that some oxygen-containing compounds can act both as acids and as bases depending on the circumstances, just as water can. Methanol and acetone, for instance, act as acids when they donate a proton but as bases when their oxygen atom accepts a proton.

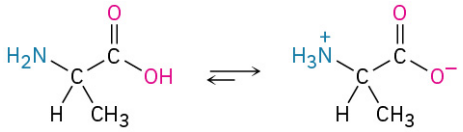


Some organic bases



Methylamine **Methanol** **Acetone**

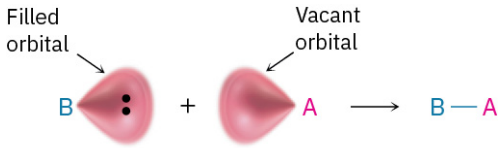
We'll see in Chapter 26 that substances called amino acids, so-named because they are both amines ($-NH_2$) and carboxylic acids ($-CO_2H$), are the building blocks from which the proteins in all living organisms are made. Twenty different amino acids go into making up proteins—alanine is an example. Interestingly, alanine and other amino acids exist primarily in a doubly charged form called a *zwitterion* rather than in the uncharged form. The zwitterion form arises because amino acids have both acidic and basic sites within the same molecule and therefore undergo an internal acid–base reaction.



Alanine (uncharged form) **Alanine (zwitterion form)**

2.11 Acids and Bases: The Lewis Definition

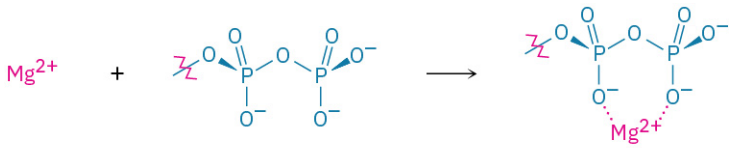
The Lewis definition of acids and bases is more encompassing than the Brønsted–Lowry definition because it's not limited to substances that donate or accept just protons. A **Lewis acid** is a substance that accepts an electron pair, and a **Lewis base** is a substance that donates an electron pair. The donated electron pair is shared between the acid and the base in a covalent bond.



Lewis base **Lewis acid**

Lewis Acids and the Curved Arrow Formalism

The fact that a Lewis acid is able to accept an electron pair means that it must have either a vacant, low-energy orbital or a polar bond to hydrogen so that it can donate H^+ (which has an empty 1*s* orbital). Thus, the Lewis definition of acidity includes many species in addition to H^+. For example, various metal cations, such as Mg^{2+}, are Lewis acids because they accept a pair of electrons when they form a bond to a base. We'll also see in later chapters that certain metabolic reactions begin with an acid–base reaction between Mg^{2+} as a Lewis acid and an organic diphosphate or triphosphate ion as the Lewis base.



Lewis acid **Lewis base (an organodiphosphate ion)** **Acid–base complex**

In the same way, compounds of group 3A elements, such as BF_3 and $AlCl_3$, are Lewis acids because they have

unfilled valence orbitals and can accept electron pairs from Lewis bases, as shown in **FIGURE 2.6**. Similarly, many transition-metal compounds, such as $TiCl_4$, $FeCl_3$, $ZnCl_2$, and $SnCl_4$, are Lewis acids.

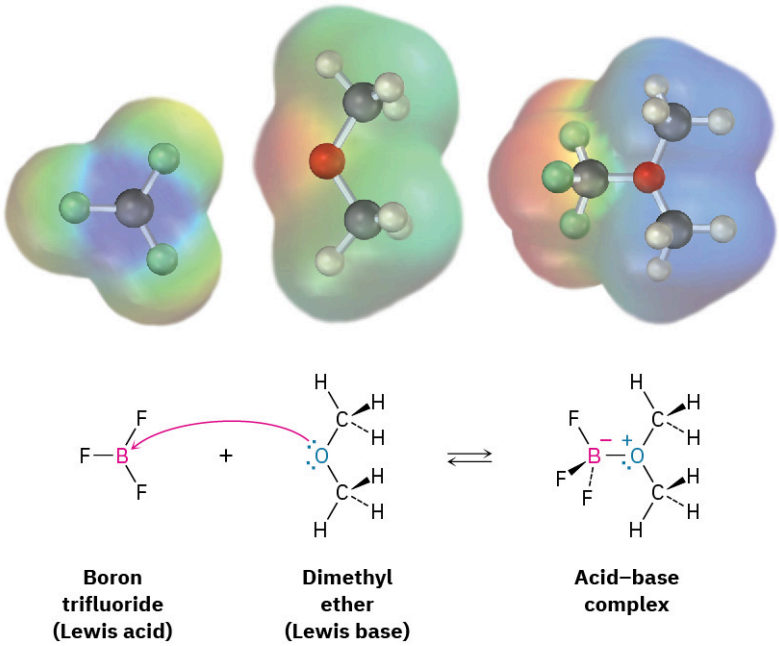


FIGURE 2.6 The reaction of boron trifluoride, a Lewis acid, with dimethyl ether, a Lewis base. The Lewis acid accepts a pair of electrons, and the Lewis base donates a pair of nonbonding electrons. Note how the movement of electrons from the Lewis base to the Lewis acid is indicated by a curved arrow. Note also how, in electrostatic potential maps, **the boron becomes more negative (red)** after reaction because it has gained electrons and **the oxygen atom becomes more positive (blue)** because it has donated electrons.

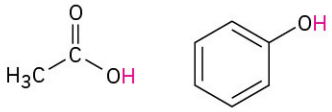
Look closely at the acid–base reaction in **FIGURE 2.6**, and notice how it's shown. Dimethyl ether, the Lewis base, donates an electron pair to a vacant valence orbital of the boron atom in BF_3, a Lewis acid. The direction of electron-pair flow from base to acid is shown using a curved arrow, just as the direction of electron flow from one resonance structure to another was shown using curved arrows in **Section 2.5**. We'll use this curved-arrow notation throughout the remainder of this text to indicate electron flow during reactions, so get used to seeing it.

Some further examples of Lewis acids follow:

Some Lewis acids

Some neutral proton donors:

H_2O HCl HBr HNO_3 H_2SO_4



A carboxylic acid

A phenol

CH_3CH_2OH

An alcohol

Some cations:

Li^+ Mg^{2+}

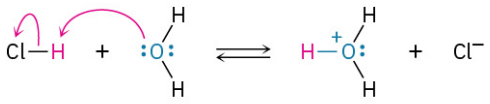
Some metal compounds:

$AlCl_3$ $TiCl_4$ $FeCl_3$ $ZnCl_2$

Lewis Bases

The Lewis definition of a base—a compound with a pair of nonbonding electrons that it can use to bond to a Lewis acid—is similar to the Brønsted–Lowry definition. Thus, H_2O, with its two pairs of nonbonding electrons

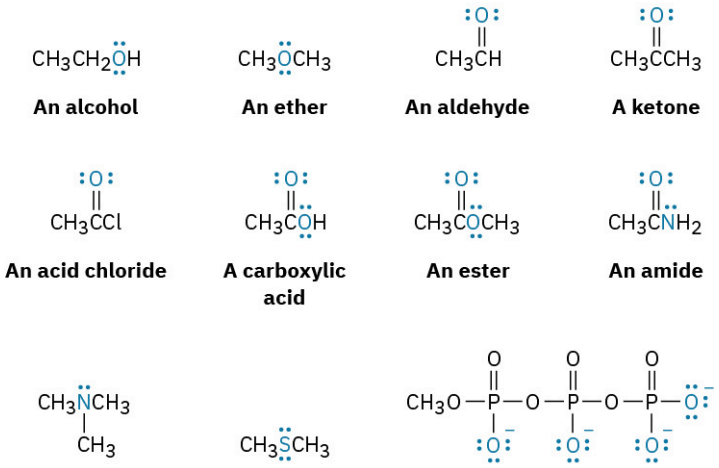
on oxygen, acts as a Lewis base by donating an electron pair to an H^+ in forming the hydronium ion, H_3O^+.



Acid **Base** **Hydronium ion**

In a more general sense, most oxygen- and nitrogen-containing organic compounds can act as Lewis bases because they have lone pairs of electrons. A divalent oxygen compound has two lone pairs of electrons, and a trivalent nitrogen compound has one lone pair. Note in the following examples that some compounds can act as both acids and bases, just as water can. Alcohols and carboxylic acids, for instance, act as acids when they donate an H^+ but as bases when their oxygen atom accepts an H^+.

Some Lewis bases

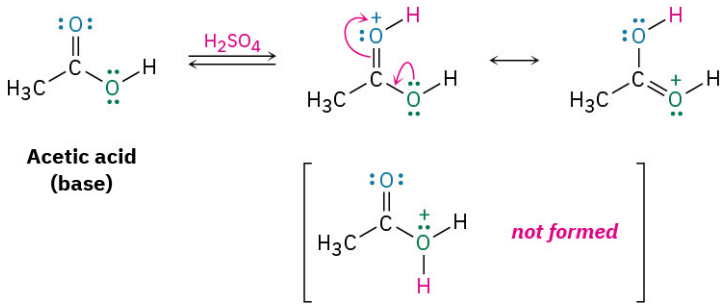


An alcohol **An ether** **An aldehyde** **A ketone**

An acid chloride **A carboxylic acid** **An ester** **An amide**

An amine **A sulfide** **An organotriphosphate ion**

Note also that some Lewis bases, such as carboxylic acids, esters, and amides, have more than one atom with a lone pair of electrons and can therefore react at more than one site. Acetic acid, for example, can be protonated either on the doubly bonded oxygen atom or on the singly bonded oxygen atom. Reaction normally occurs only once in such instances, and the more stable of the two possible protonation products is formed. For acetic acid, protonation by reaction with sulfuric acid occurs on the doubly bonded oxygen because that product is stabilized by two resonance forms.



Acetic acid (base)

WORKED EXAMPLE 2.6

Using Curved Arrows to Show Electron Flow

Using curved arrows, show how acetaldehyde, CH_3CHO, can act as a Lewis base.

Strategy

A Lewis base donates an electron pair to a Lewis acid. We therefore need to locate the electron lone pairs on acetaldehyde and use a curved arrow to show the movement of a pair toward the H atom of the acid.

Solution

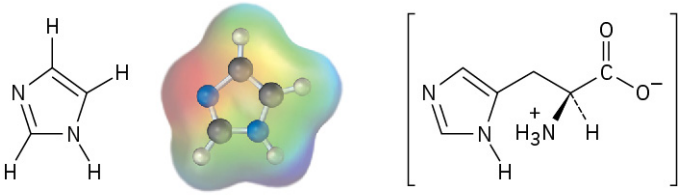


Acetaldehyde

PROBLEM 2-17 Using curved arrows, show how the species in part **(a)** can act as Lewis bases in their reactions with HCl, and show how the species in part **(b)** can act as Lewis acids in their reaction with OH^-.

(a) CH_3CH_2OH, $HN(CH_3)_2$, $P(CH_3)_3$ **(b)** H_3C^+, $B(CH_3)_3$, $MgBr_2$

PROBLEM 2-18 Imidazole, which forms part of amino acid histidine, can act as both an acid and a base.



Imidazole Histidine

(a) Look at the electrostatic potential map of imidazole, and identify the most acidic hydrogen atom and the most basic nitrogen atom.

(b) Draw structures for the resonance forms of the products that result when imidazole is protonated by an acid and deprotonated by a base.

2.12 Noncovalent Interactions between Molecules

When thinking about chemical reactivity, chemists usually focus their attention on bonds, the covalent interactions between atoms within molecules. Also important, however, particularly in large biomolecules like proteins and nucleic acids, are a variety of interactions *between* molecules that strongly affect molecular properties. Collectively called either **intermolecular forces**, **van der Waals forces**, or **noncovalent interactions**, they are of several different types: dipole–dipole forces, dispersion forces, and hydrogen bonds.

Dipole–dipole forces occur between polar molecules as a result of electrostatic interactions among dipoles. The forces can be either attractive or repulsive depending on the orientation of the molecules—attractive when unlike charges are together and repulsive when like charges are together. The attractive geometry is lower in energy and therefore predominates (FIGURE 2.7).

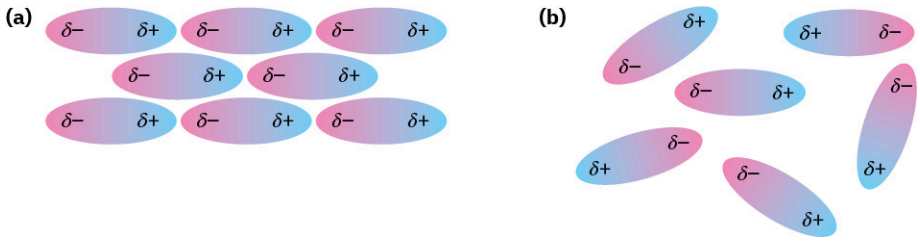


FIGURE 2.7 **Dipole–dipole forces cause polar molecules (a) to attract one another when they orient with unlike charges together, but (b) to repel one another when they orient with like charges together.**

Dispersion forces occur between all neighboring molecules and arise because the electron distribution within molecules is constantly changing. Although uniform on a time-averaged basis, the electron distribution even in nonpolar molecules is likely to be nonuniform at any given instant. One side of a molecule may, by chance, have a slight excess of electrons relative to the opposite side, giving the molecule a temporary dipole. This temporary dipole in one molecule causes a nearby molecule to adopt a temporarily opposite dipole, resulting in a tiny attraction between the two (FIGURE 2.8). Temporary molecular dipoles have only a fleeting existence and are constantly changing, but their cumulative effect is often strong enough to hold molecules close together so that a substance is a liquid or solid rather than a gas.

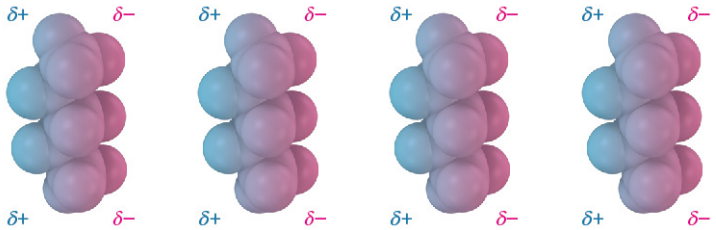
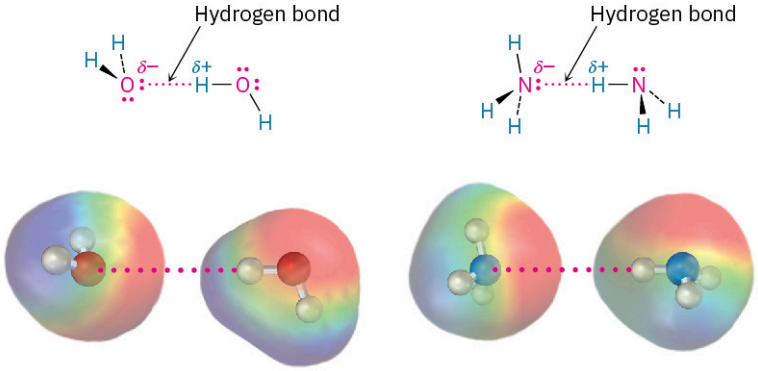
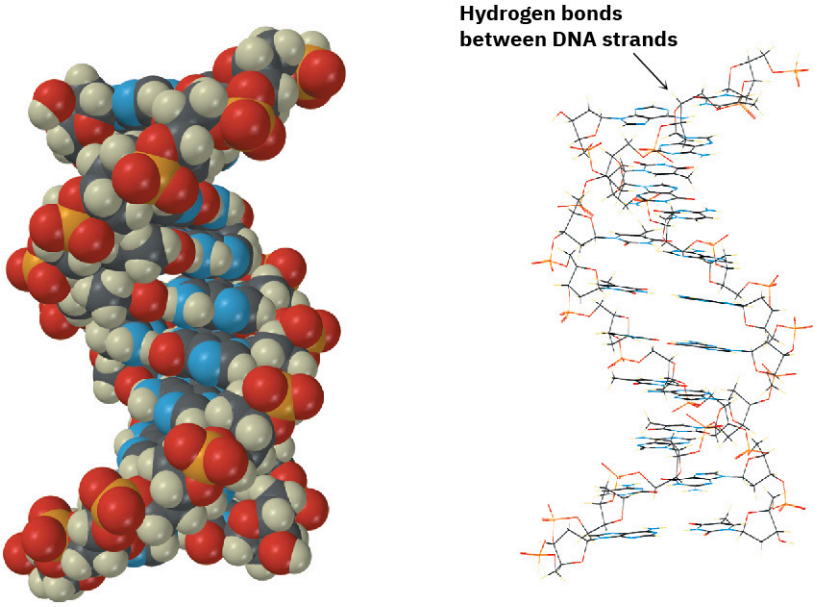


FIGURE 2.8 Attractive dispersion forces in nonpolar molecules are caused by temporary dipoles, as shown in these models of pentane, C_5H_{12}.

Perhaps the most important noncovalent interaction in biological molecules is the **hydrogen bond**, an attractive interaction between a hydrogen atom bonded to an electronegative O or N atom and an unshared electron pair on another O or N atom. In essence, a hydrogen bond is a very strong dipole–dipole interaction involving polarized O–H or N–H bonds. Electrostatic potential maps of water and ammonia clearly show the positively polarized hydrogens (blue) and the negatively polarized oxygens and nitrogens (red).



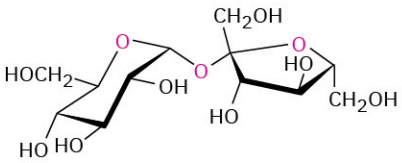
Hydrogen bonding has enormous consequences for living organisms. Hydrogen bonds cause water to be a liquid rather than a gas at ordinary temperatures, they hold enzymes in the shapes necessary for catalyzing biological reactions, and they cause strands of deoxyribonucleic acid (DNA) to pair up and coil into the double helix that stores genetic information.



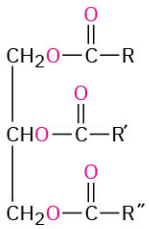
A deoxyribonucleic acid segment

One further point before leaving the subject of noncovalent interactions: biochemists frequently use the term **hydrophilic**, meaning "water-loving," to describe a substance that is attracted to water and the term **hydrophobic**, meaning "water-fearing," to describe a substance that is not strongly attracted to water.

Hydrophilic substances, such as table sugar, often have a number of –OH groups in their structure so they can form hydrogen bonds and dissolve in water, whereas hydrophobic substances, such as vegetable oil, do not have groups that form hydrogen bonds and do not dissolve in water.

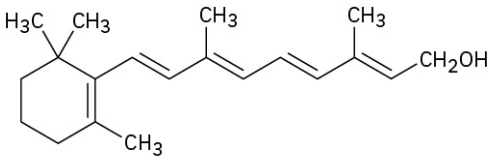


Sucrose
(a disaccharide)

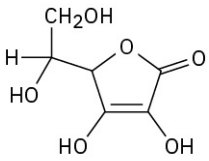


Animal fat—a triester
(R, R′, R″ = C_{11}—C_{19} chains)

PROBLEM 2-19 Of the two vitamins A and C, one is hydrophilic and water-soluble while the other is hydrophobic and fat-soluble. Which is which?



Vitamin A
(retinol)



Vitamin C
(ascorbic acid)

CHEMISTRY MATTERS

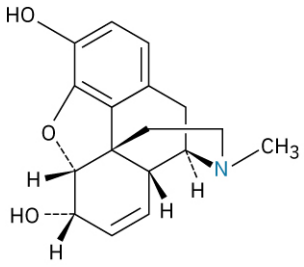
Alkaloids: From Cocaine to Dental Anesthetics

Just as ammonia (NH_3) is a weak base, there are a large number of nitrogen-containing organic compounds called amines that are also weak bases. In the early days of organic chemistry, basic amines derived from natural sources were known as vegetable alkali, but they are now called **alkaloids**. More than 20,000 alkaloids are known. Their study provided much of the impetus for the growth of organic chemistry in the nineteenth century and remains today an active and fascinating area of research.

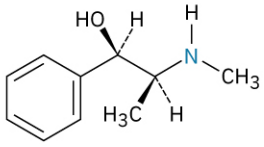


FIGURE 2.9 The coca bush *Erythroxylon coca,* native to upland rain forest areas of Colombia, Ecuador, Peru, Bolivia, and western Brazil, is the source of the alkaloid cocaine. (credit: "*Erythroxylum coca*" by Danna Guevara/Wikimedia Commons, CC BY 4.0)

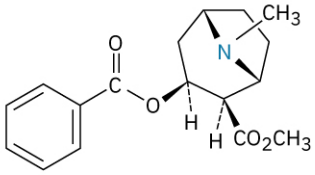
Many alkaloids have pronounced biological properties, and approximately 50% of pharmaceutical agents used today are derived from naturally occurring amines. As just three examples, morphine, an analgesic agent (painkiller), is obtained from the opium poppy *Papaver somniferum.* Ephedrine, a bronchodilator, decongestant, and appetite suppressant, is obtained from *Ephedra sinica*, an evergreen shrub native to Mongolia and northeastern China. Cocaine, both an anesthetic and a stimulant, is obtained from the coca bush *Erythroxylon coca,* endemic to the upland rain forest areas of central South America. (And yes, there really was a small amount of cocaine in the original Coca-Cola recipe, although it was removed in 1906.)



Morphine

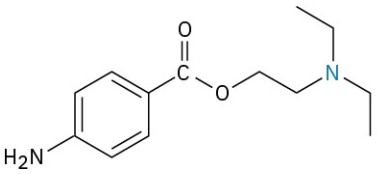


Ephedrine

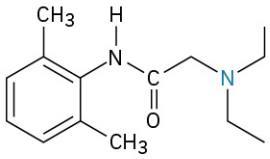


Cocaine

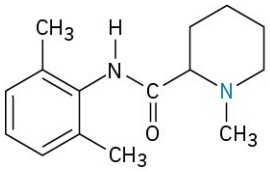
Cocaine itself is rarely used medically because it is too addictive, but its anesthetic properties provoked a long search for related but nonaddictive compounds. This search ultimately resulted in the synthesis of the "caine" anesthetics that are commonly used today in dental and surgical anesthesia. Procaine, the first such compound, was synthesized in 1898 and marketed under the name Novocain. It was rapidly adopted and remains in use today as a topical anesthetic. Other related compounds with different activity profiles followed: Lidocaine, marketed as Xylocaine, was introduced in 1943, and mepivacaine (Carbocaine) in the early 1960s. More recently, bupivacaine (Marcaine) and prilocaine (Citanest) have gained popularity. Both are quick-acting, but the effects of bupivacaine last for 3 to 6 hours while those of prilocaine fade after 45 minutes. Notice some structural similarity of all the caines to cocaine itself.



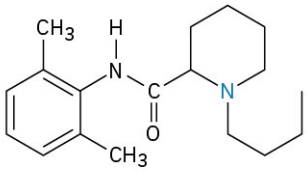
Procaine
(Novocain)



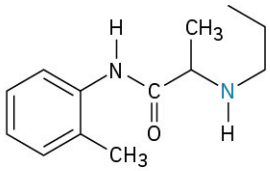
Lidocaine
(Xylocaine)



Mepivacaine
(Carbocaine)



Bupivacaine
(Marcaine)



Prilocaine
(Citanest)

An estimate from the U.S. National Academy of Sciences is that less than 1% of all living species have been characterized. Thus, alkaloid chemistry remains an active area of research, and innumerable substances with potentially useful properties have yet to be discovered. Undoubtedly even the caine anesthetics will become obsolete at some point, perhaps supplanted by newly discovered alkaloids.

Key Terms

- **acidity constant (K_a)**
- **alkaloid**
- **Brønsted–Lowry acid**
- **Brønsted–Lowry base**
- **conjugate acid**
- **conjugate base**
- **dipole moment (μ)**
- **dispersion force**
- **electronegativity (EN)**
- **electrostatic potential map**
- **formal charge**
- **hydrogen bond**
- **hydrophilic**
- **hydrophobic**
- **inductive effect**
- **intermolecular force**
- **Lewis acid**
- **Lewis base**
- **noncovalent interaction**
- **pK_a**
- **polar covalent bond**
- **resonance form**
- **resonance hybrid**
- **van der Waals force**

Summary

Understanding organic chemistry means knowing not just what happens but also why and how it happens at the molecular level. In this chapter, we've reviewed some of the ways that chemists describe and account for chemical reactivity, thereby providing a foundation for understanding the specific reactions that will be discussed in subsequent chapters.

Organic molecules often have **polar covalent bonds** as a result of unsymmetrical electron sharing caused by differences in the **electronegativity** of atoms. A carbon–oxygen bond is polar, for example, because oxygen attracts the shared electrons more strongly than carbon does. Carbon–hydrogen bonds are relatively nonpolar. Many molecules as a whole are also polar, owing to the presence of individual polar bonds and electron lone pairs. The polarity of a molecule is measured by its **dipole moment**, $\boldsymbol{\mu}$.

Plus (+) and minus (–) signs are often used to indicate the presence of **formal charges** on atoms in molecules. Assigning formal charges to specific atoms is a bookkeeping technique that makes it possible to keep track of the valence electrons around an atom and offers some clues about chemical reactivity.

Some substances, such as acetate ion and benzene, can't be represented by a single line-bond structure and must be considered as a **resonance hybrid** of two or more structures, none of which would be correct by themselves. The only difference between two **resonance forms** is in the location of their π and nonbonding electrons. The nuclei remain in the same places in both structures, and the hybridization of the atoms remains

the same.

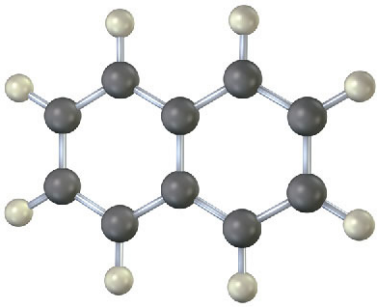
Acidity and basicity are closely related to the ideas of polarity and electronegativity. A **Brønsted–Lowry acid** is a compound that can donate a proton (hydrogen ion, H^+), and a **Brønsted–Lowry base** is a compound that can accept a proton. The strength of a Brønsted–Lowry acid or base is expressed by its **acidity constant, K_a**, or by the negative logarithm of the acidity constant, **pK_a**. The larger the pK_a, the weaker the acid. More useful is the Lewis definition of acids and bases. A **Lewis acid** is a compound that has a low-energy empty orbital that can accept an electron pair; Mg^{2+}, BF_3, $AlCl_3$, and H^+ are examples. A **Lewis base** is a compound that can donate an unshared electron pair; NH_3 and H_2O are examples. Most organic molecules that contain oxygen and nitrogen can act as Lewis bases toward sufficiently strong acids.

A variety of **noncovalent interactions** have a significant effect on the properties of large biomolecules. **Hydrogen bonding**—the attractive interaction between a positively polarized hydrogen atom bonded to an oxygen or nitrogen atom with an unshared electron pair on another O or N atom, is particularly important in giving proteins and nucleic acids their shapes.

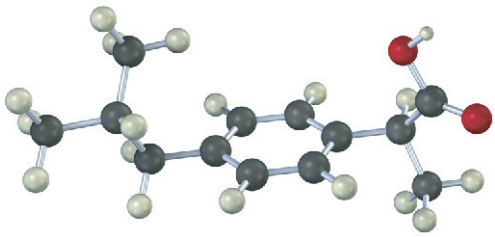
Additional Problems

Visualizing Chemistry

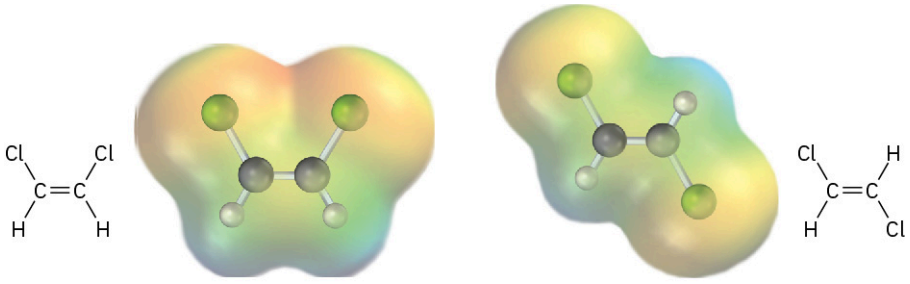
PROBLEM 2-20 Fill in the multiple bonds in the following model of naphthalene, $C_{10}H_8$ (black = C, gray = H). How many resonance structures does naphthalene have? Draw them.



PROBLEM 2-21 The following model is a representation of ibuprofen, a common over-the-counter pain reliever. Indicate the positions of the multiple bonds, and draw a skeletal structure (black = C, red = O, gray = H).



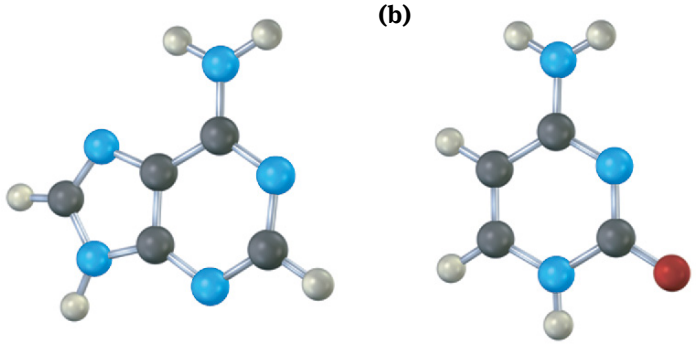
PROBLEM 2-22 *cis*-1,2-Dichloroethylene and *trans*-1,2-dichloroethylene are *isomers,* compounds with the same formula but different chemical structures. Look at the following electrostatic potential maps, and tell whether either compound has a dipole moment.



***cis*-1,2-Dichloroethylene** ***trans*-1,2-Dichloroethylene**

PROBLEM 2-23 The following molecular models are representations of **(a)** adenine and **(b)** cytosine, constituents of DNA (deoxyribonucleic acid). Indicate the positions of multiple bonds and lone pairs for both, and draw skeletal structures (black = C, red = O, blue = N, gray = H).

(a) **(b)**

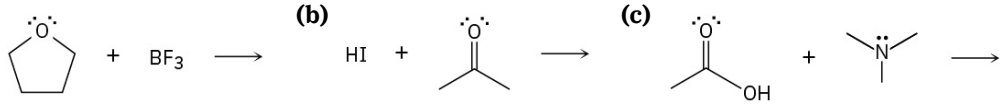


Adenine **Cytosine**

Mechanism Problems

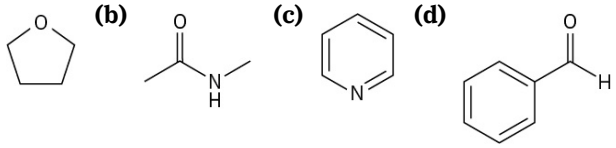
PROBLEM 2-24 Predict the product(s) of the following acid/base reactions. Draw curved arrows to show the formation and breaking of bonds.

(a) **(b)** **(c)**



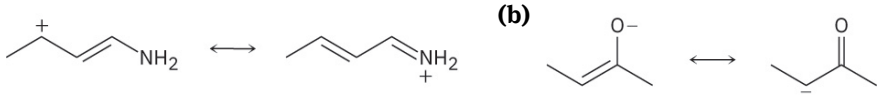
PROBLEM 2-25 Use curved arrows to draw the protonated form of the following Lewis bases.

(a) **(b)** **(c)** **(d)**

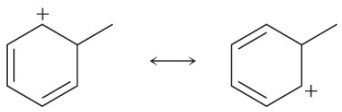


PROBLEM 2-26 Use the curved-arrow formalism to show how the electrons flow in the resonance form on the left to give the one on the right.

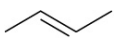
(a) **(b)**



(c)



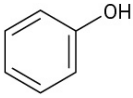
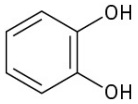
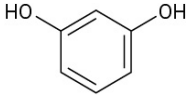
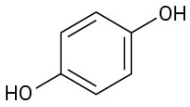
PROBLEM 2-27 Double bonds can also act like Lewis bases, sharing their electrons with Lewis acids. Use curved arrows to show how each of the following double bonds will react with HCl and draw the resulting carbocation.

(a) $H_2C=CH_2$ **(b)**  **(c)** 

Electronegativity and Dipole Moments

PROBLEM 2-28 Identify the most electronegative element in each of the following molecules:
(a) CH_2FCl **(b)** $FCH_2CH_2CH_2Br$ **(c)** $HOCH_2CH_2NH_2$ **(d)** CH_3OCH_2Li

PROBLEM 2-29 Use the electronegativity table given in Figure 2.3 to predict which bond in each of the following pairs is more polar, and indicate the direction of bond polarity for each compound.
(a) H_3C–Cl or Cl–Cl **(b)** H_3C–H or H–Cl **(c)** HO–CH_3 or $(CH_3)_3Si$–CH_3 **(d)** H_3C–Li or Li–OH

PROBLEM 2-30 Which of the following molecules has a dipole moment? Indicate the expected direction of each.
(a)  **(b)**  **(c)**  **(d)** 

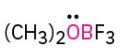
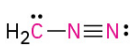
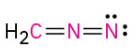
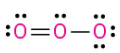
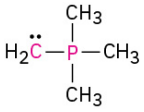
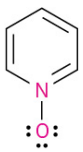
PROBLEM 2-31 **(a)** The H–Cl bond length is 136 pm. What would the dipole moment of HCl be if the molecule were 100% ionic, $H^+ Cl^-$?
(b) The actual dipole moment of HCl is 1.08 D. What is the percent ionic character of the H–Cl bond?

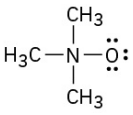
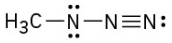
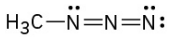
PROBLEM 2-32 Phosgene, $Cl_2C{=}O$, has a smaller dipole moment than formaldehyde, $H_2C{=}O$, even though it contains electronegative chlorine atoms in place of hydrogen. Explain.

PROBLEM 2-33 Fluoromethane (CH_3F, μ = 1.81 D) has a smaller dipole moment than chloromethane (CH_3Cl, μ = 1.87 D) even though fluorine is more electronegative than chlorine. Explain.

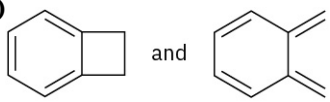
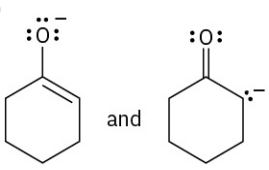
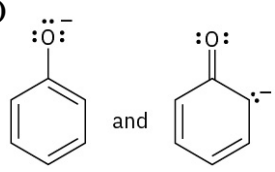
PROBLEM 2-34 Methanethiol, CH_3SH, has a substantial dipole moment (μ = 1.52) even though carbon and sulfur have identical electronegativities. Explain.

Formal Charges

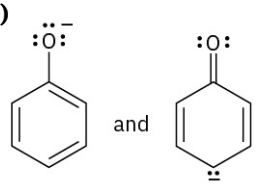
PROBLEM 2-35 Calculate the formal charges on the atoms shown in red.
(a)  **(b)**  **(c)**  **(d)**  **(e)**  **(f)** 

PROBLEM 2-36 Assign formal charges to the atoms in each of the following molecules:
(a)  **(b)**  **(c)** 

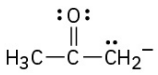
Resonance

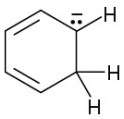
PROBLEM 2-37 Which of the following pairs of structures represent resonance forms?
(a)  **(b)**  **(c)** 

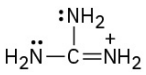
(d)

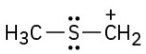


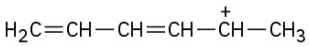
PROBLEM 2-38 Draw as many resonance structures as you can for the following species:

(a) 

(b) 

(c) 

(d) 

(e) 

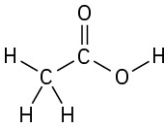
PROBLEM 2-39 1,3-Cyclobutadiene is a rectangular molecule with two shorter double bonds and two longer single bonds. Why do the following structures not represent resonance forms?



Acids and Bases

PROBLEM 2-40 Alcohols can act either as weak acids or as weak bases, just as water can. Show the reaction of methanol, CH_3OH, with a strong acid such as HCl and with a strong base such as $Na^+ {}^-NH_2$

PROBLEM 2-41 The O–H hydrogen in acetic acid is more acidic than any of the C–H hydrogens. Explain this result using resonance structures.

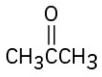


Acetic acid

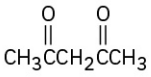
PROBLEM 2-42 Draw electron-dot structures for the following molecules, indicating any unshared electron pairs. Which of the compounds are likely to act as Lewis acids and which as Lewis bases?
(a) $AlBr_3$ **(b)** $CH_3CH_2NH_2$ **(c)** BH_3 **(d)** HF **(e)** CH_3SCH_3 **(f)** $TiCl_4$

PROBLEM 2-43 Write the products of the following acid–base reactions:
(a) $CH_3OH + H_2SO_4 \rightleftarrows$? **(b)** $CH_3OH + NaNH_2 \rightleftarrows$? **(c)** $CH_3NH_3^+ Cl^- + NaOH \rightleftarrows$?

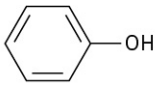
PROBLEM 2-44 Rank the following substances in order of increasing acidity:



Acetone
(pK_a = 19.3)



2,4-Pentanedione
(pK_a = 9)



Phenol
(pK_a = 9.9)

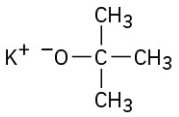


Acetic acid
(pK_a = 4.76)

PROBLEM 2-45 Which, if any, of the substances in Problem 2-44 is a strong enough acid to react almost completely with NaOH? (The pK_a of H_2O is 15.74.)

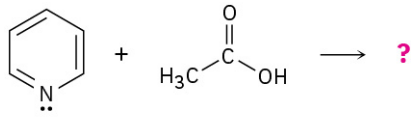
PROBLEM 2-46 The ammonium ion (NH_4^+, pK_a = 9.25) has a lower pK_a than the methylammonium ion ($CH_3NH_3^+$, pK_a = 10.66). Which is the stronger base, ammonia (NH_3) or methylamine (CH_3NH_2)? Explain.

PROBLEM 2-47 Is *tert*-butoxide anion a strong enough base to react significantly with water? In other words, can a solution of potassium *tert*-butoxide be prepared in water? The pK_a of *tert*-butyl alcohol is approximately 18.



Potassium *tert*-butoxide

PROBLEM 2-48 Predict the structure of the product formed in the reaction of the organic base pyridine with the organic acid acetic acid, and use curved arrows to indicate the direction of electron flow.



Pyridine **Acetic acid**

PROBLEM 2-49 Calculate K_a values from the following pK_a's:
(a) Acetone, pK_a = 19.3 **(b)** Formic acid, pK_a = 3.75

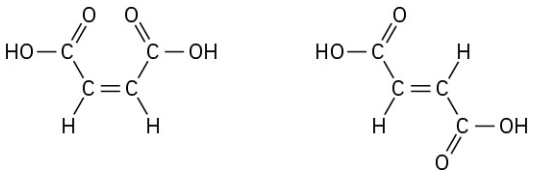
PROBLEM 2-50 Calculate pK_a values from the following K_a's:
(a) Nitromethane, $K_a = 5.0 \times 10^{-11}$ **(b)** Acrylic acid, $K_a = 5.6 \times 10^{-5}$

PROBLEM 2-51 What is the pH of a 0.050 M solution of formic acid, pK_a = 3.75?

PROBLEM 2-52 Sodium bicarbonate, $NaHCO_3$, is the sodium salt of carbonic acid (H_2CO_3), pK_a = 6.37. Which of the substances shown in Problem 2-44 will react significantly with sodium bicarbonate?

General Problems

PROBLEM 2-53 Maleic acid has a dipole moment, but the closely related fumaric acid, a substance involved in the citric acid cycle by which food molecules are metabolized, does not. Explain.

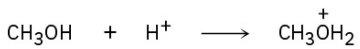


Maleic acid **Fumaric acid**

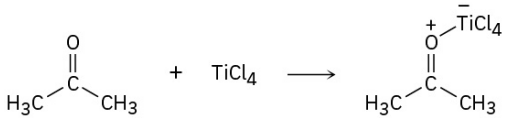
PROBLEM 2-54 Assume that you have two unlabeled bottles, one of which contains phenol (pK_a = 9.9) and one of which contains acetic acid (pK_a = 4.76). In light of your answer to Problem 2-52, suggest a simple way to determine what is in each bottle.

PROBLEM 2-55 Identify the acids and bases in the following reactions:

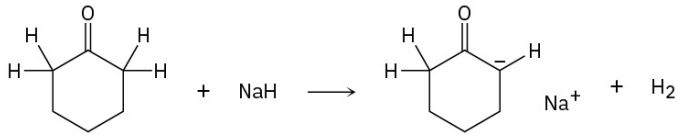
(a)



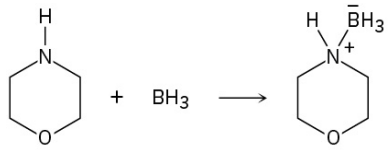
(b)



(c)

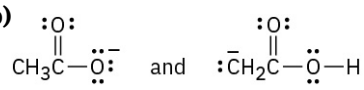


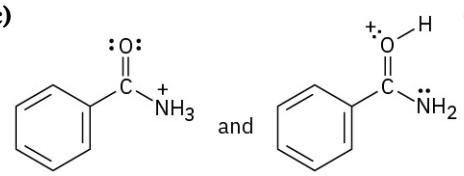
(d)

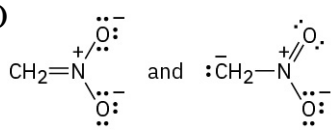


PROBLEM 2-56 Which of the following pairs represent resonance structures?

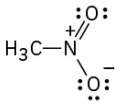
(a)  and

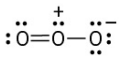
(b) 

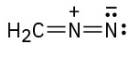
(c) 

(d) 

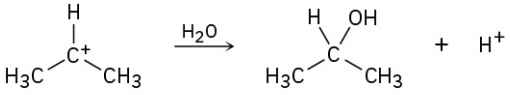
PROBLEM 2-57 Draw as many resonance structures as you can for the following species, adding appropriate formal charges to each:

(a) Nitromethane, 

(b) Ozone, 

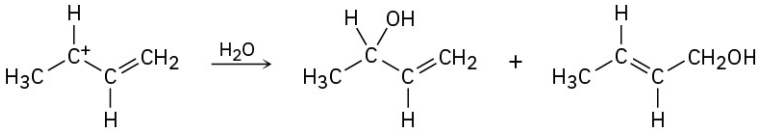
(c) Diazomethane, 

PROBLEM 2-58 Carbocations, which contain a trivalent, positively charged carbon atom, react with water to give alcohols:



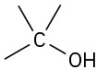
A carbocation **An alcohol**

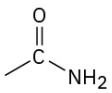
How can you account for the fact that the following carbocation gives a mixture of *two* alcohols on reaction with water?



PROBLEM 2-59 We'll see in the next chapter that organic molecules can be classified according to the functional groups they contain, where a functional group is a collection of atoms with a characteristic chemical reactivity. Use the electronegativity values given in Figure 2.3 to predict the direction of polarization of the following functional groups.

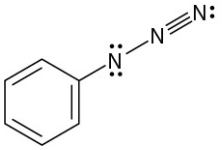
(a)  **Ketone**

(b)  **Alcohol**

(c)  **Amide**

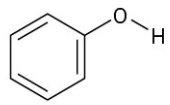
(d)  **Nitrile**

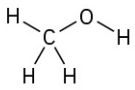
PROBLEM 2-60 The *azide* functional group, which occurs in azidobenzene, contains three adjacent nitrogen atoms. One resonance structure for azidobenzene is shown. Draw three additional resonance structures, and assign appropriate formal charges to the atoms in all four.



Azidobenzene

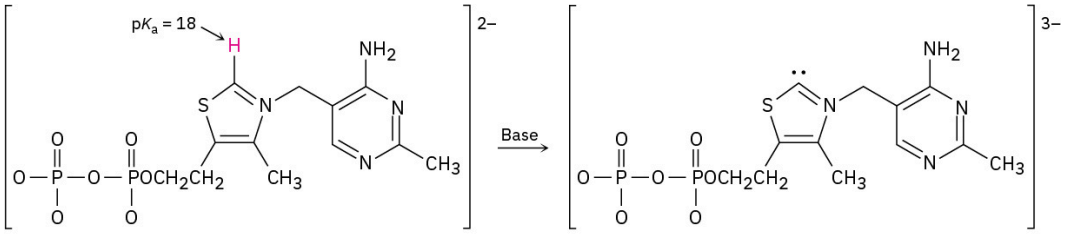
PROBLEM 2-61 Phenol, C_6H_5OH, is a stronger acid than methanol, CH_3OH, even though both contain an O–H bond. Draw the structures of the anions resulting from loss of H^+ from phenol and methanol, and use resonance structures to explain the difference in acidity.





Phenol (pK_a = 9.89) **Methanol (pK_a = 15.54)**

PROBLEM 2-62 Thiamin diphosphate (TPP), a derivative of vitamin B_1 required for glucose metabolism, is a weak acid that can be deprotonated by a base. Assign formal charges to the appropriate atoms in both TPP and its deprotonation product.



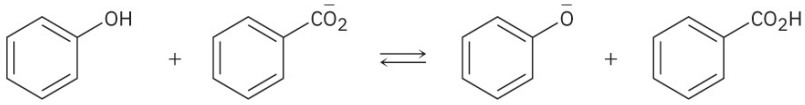
Thiamin diphosphate (TPP)

PROBLEM 2-63 Which of the following compounds or ions have a dipole moment?

(a) Carbonate ion (CO_3^{2-}) **(b)**  **(c)** 

PROBLEM 2-64 Use the pK_a table in Appendix B to determine in which direction the equilibrium is favored.

(a)



(b)



(c)



PROBLEM 2-65 Which intermolecular force is predominantly responsible for each observation below?

(a) $CH_3(CH_2)_{29}CH_3$, a component found in paraffin wax, is a solid at room temperature while $CH_3(CH_2)_6CH_3$ is a liquid.

(b) $CH_3CH_2CH_2OH$ has a higher boiling point than CH_4.

(c) CH_3CO_2H, which is found in vinegar, will dissolve in water but not in oil. Assume that oil is $CH_3(CH_2)_4CH_3$.